Histopathologic Cancer Detection: Introduction and Agenda

Using AI-Powered Solution

Introduction

Metastatic cancer, where cancer cells spread to other parts of the body, is a significant cause of cancer-related deaths globally. Early and accurate detection is crucial for improving patient outcomes and guiding treatment strategies. However, the manual examination of digital pathology images is both time-consuming and prone to human error.

To address this, our project aims to develop an automated, AI-powered solution that can accurately identify metastatic cancer from digital pathology images. Using the given benchmark dataset, we will build and deploy a deep learning model capable of classifying small image patches as cancerous or non-cancerous. This solution has the potential to enhance the efficiency and accuracy of cancer diagnoses, thereby improving patient care.

Agenda

Project Overview:

- Brief introduction to the PatchCamelyon (PCam) dataset and its relevance in cancer detection.
- Explanation of the core objective: developing and deploying two deep learning models for classifying metastatic cancer in digital pathology images.

Model Architecture 1:

- Introducing convolutional neural networks (CNNs) and why they are effective for image classification tasks.
- Overview of the selected model architecture and the training process.

Model Architecture 2:

- Introducing the pre-trained vision transformer (ViT) model and why it is considered for image classification.
- Overview of the selected model architecture and the training process.

Performance Metrics and Evaluation:

- Description of the key performance metrics (e.g., precision-recall curve, ROC-AUC, confusion matrix) used to evaluate the model.
- Discussing the methods for optimizing the model's performance.

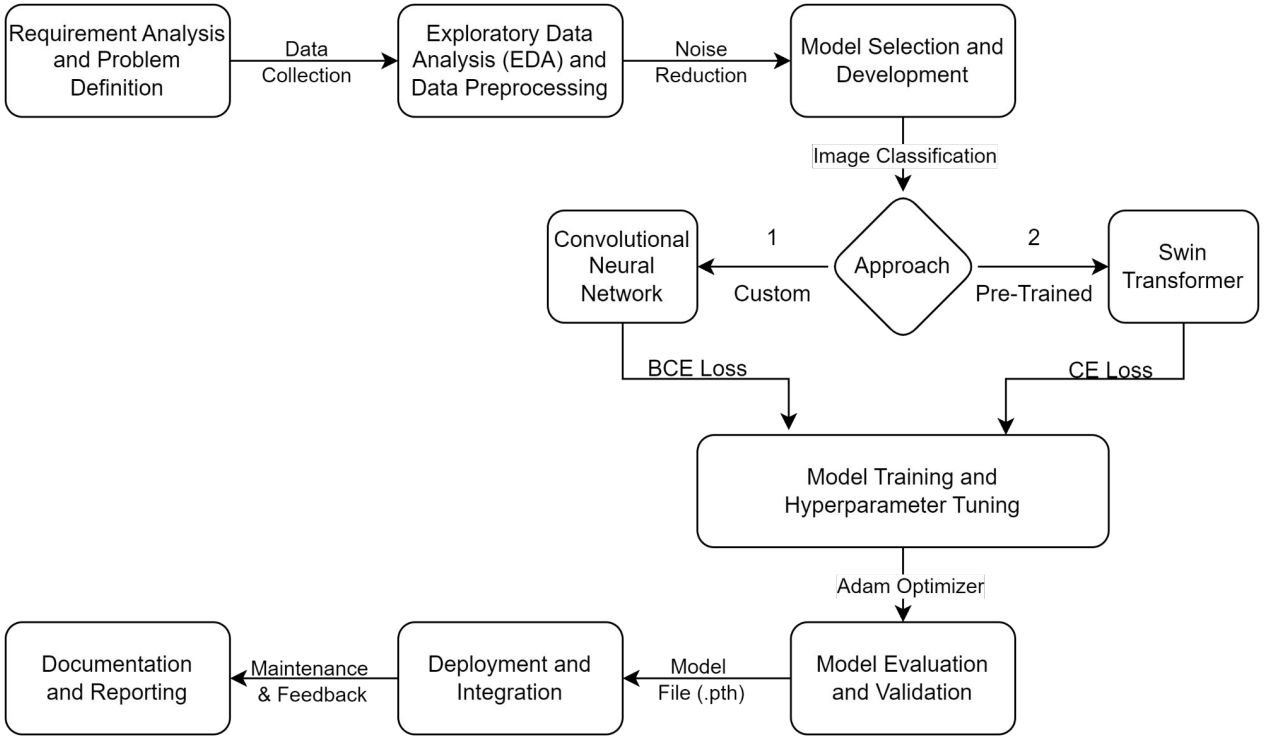
Deployment Strategy:

- How the project will be deployed on a local server and made available through a web platform locally.
- Overview of the user interface and how end-users can interact with the system.

Project Impact and Future Work:

- Highlighting the real-world impact of the project on cancer diagnosis.
- Discussing possible extensions, such as using ensemble models, Explainable AI, and real-time deployment in clinical settings.

End-to-End Architecture



Histopathologic Cancer Detection: Scope and Business Integration

Automating Cancer Detection in Pathology

Project Overview

- **Objective:** Automate the identification of metastatic cancer in pathology images.
- **Data Source:** PatchCamelyon (PCam) dataset, containing small image patches of digital pathology scans.
- **Approach:** Binary classification task using Deep learning models to distinguish between cancerous and non-cancerous tissue.

Business Objective And Market Need

- Business Objective:
  - Improve diagnostic efficiency in healthcare.
  - Reduce manual workload for pathologists.
  - Provide faster, more reliable cancer detection.
  - Enable scalability for high-volume clinical use.
- Market Need:
  - Growing demand for AI-based diagnostics in healthcare.
  - Addressing the shortage of trained pathologists.
  - Reducing human error in cancer detection.
  - Increasing accuracy in metastatic cancer diagnosis.

Interface to Existing Systems

- **Integration with Hospital Information Systems (HIS):** The model outputs can be integrated into existing HIS platforms for seamless use.
- **Interoperability with Electronic Medical Records (EMR):** Results can be added directly into patient EMR, streamlining the reporting process.
- **Data Flow and Storage:** The solution is designed to work with existing data storage systems, ensuring data compliance.
- **API Connectivity:** The system can interface with existing diagnostic tools and imaging software via APIs for real-time analysis.

Technical Feasibility and Future Integration

- Technical Feasibility
  - Trained Model can be made widely available on cloud platforms (e.g., AWS, Azure).
  - Single-GPU support for fast training and deployment.
  - Can be scaled for larger datasets without requiring extensive hardware.
  - Easily deployable in clinical settings.
- Future Integration
  - Expanding to other types of cancer detection (e.g., skin, lung).
  - Incorporating clinical data for enhanced predictions.
  - Leveraging federated learning to train models across institutions.
  - Enhancing explainability to increase trust in AI-based diagnostics.

Histopathologic Cancer Detection: Methods and Achievements

Approach 1
Custom CNN Model

Data Subsets

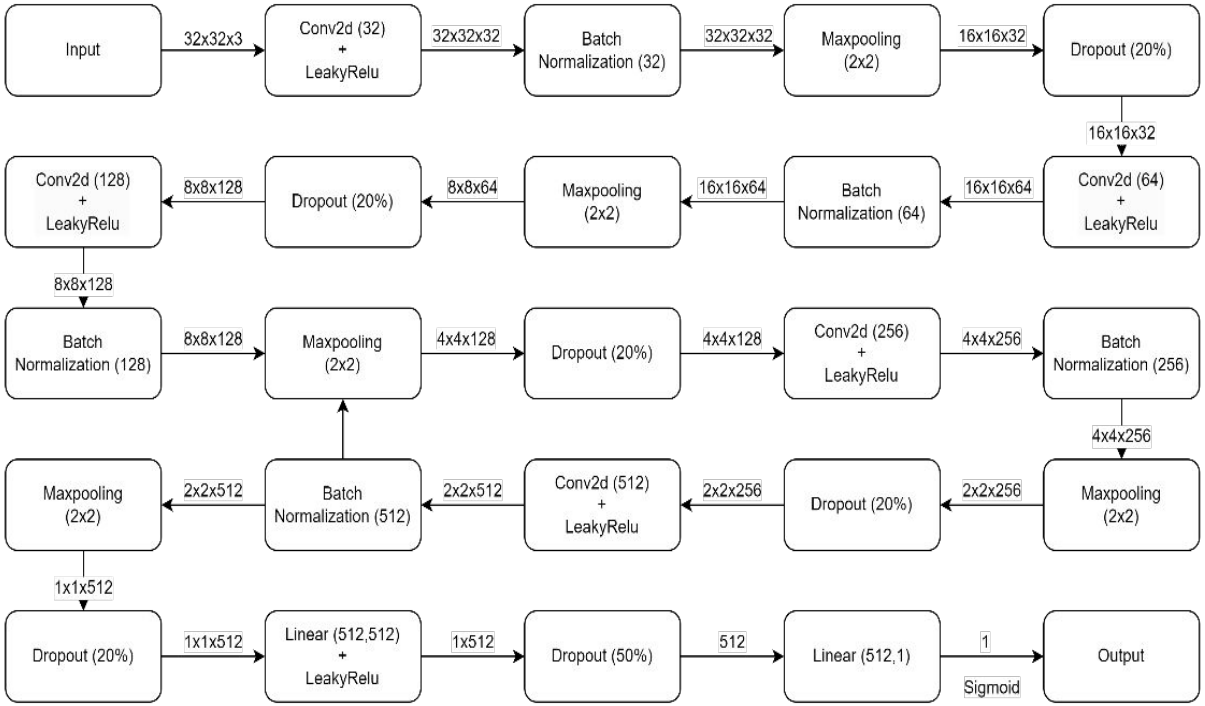
Subset 1	Cancerous	Non-Cancerous	Total
Train	16503	20896	37399
Validation	660	2640	3300
Test	660	2640	3300

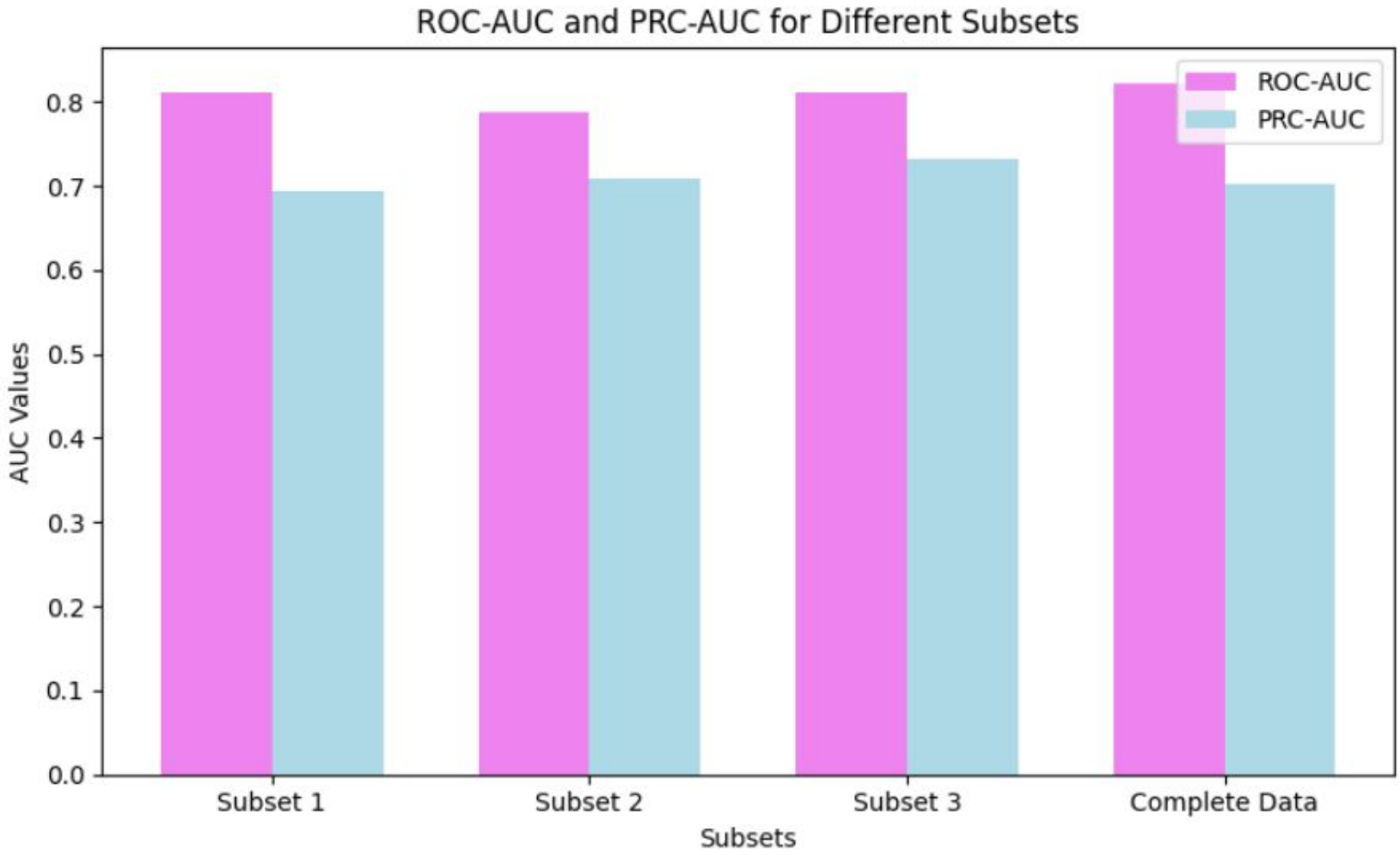
Subset 2	Cancerous	Non-Cancerous	Total
Train	33007	41791	74798
Test	1320	5280	6600
Validation	1320	5280	6600

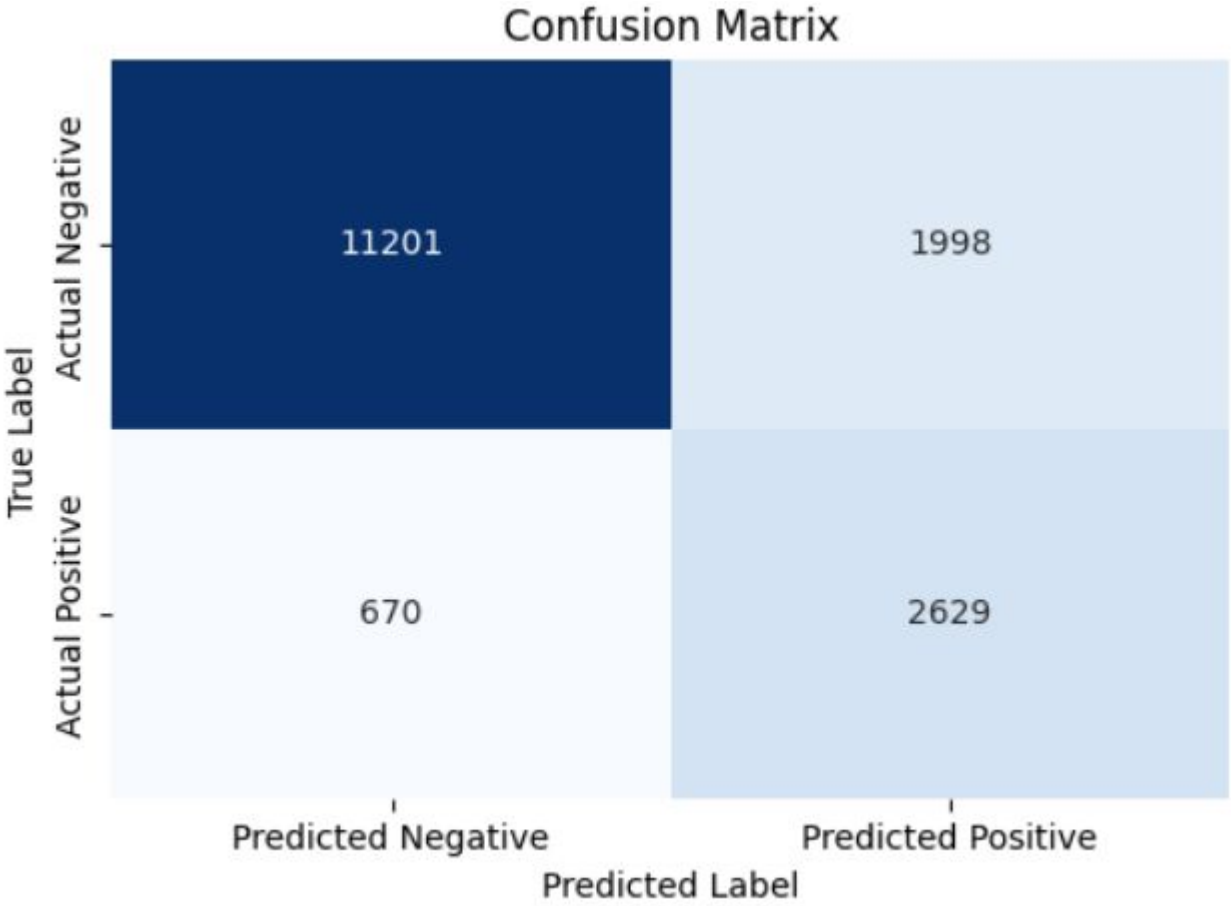
Subset 3	Cancerous	Non-Cancerous	Total
Train	49510	62687	112197
Validation	1980	7920	9900
Test	1980	7920	9900

Complete Data	Cancerous	Non-Cancerous	Total
Train	82519	104481	187000
Test	3299	13199	16498
Validation	3299	13199	16498

Model Architecture




ROC-AUC and PRC-AUC for Different Subsets
ROC-AUC
PRC-AUC
AUC Values
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Subset 1
Subset 2
Subset 3
Complete Data
Subsets


Confusion Matrix
True Label
Actual Negative
Actual Positive
11201
1998
670
2629
Predicted Negative
Predicted Positive
Predicted Label

Histopathologic Cancer Detection: Methods and Achievements

Approach 2
Pretrained ViT Model

Technical Overview:

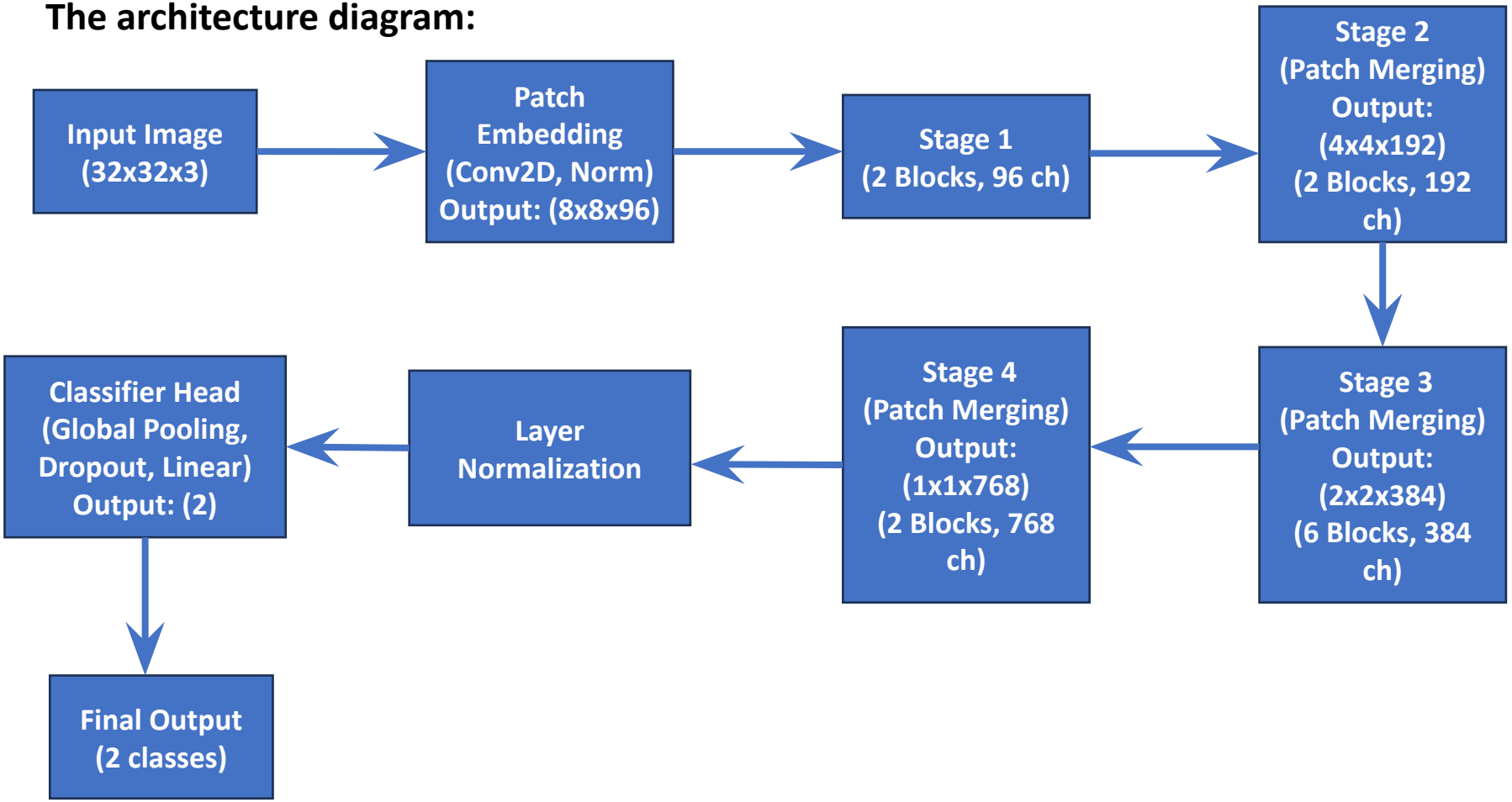
- **Framework**: PyTorch and Torchvision, using Swin Transformer for image classification.
- **Transformer Architecture**: Swin Transformer enables complex image feature extraction with global context awareness.
- **Dataset**: Histopathologic Cancer Detection dataset with labelled image data.

Methods and Data Used

Content:

- Model: Swin Transformer
- Input: 32x32 images, divided into 8x8 patches (patch size 4)
- Hierarchical feature extraction with window-based attention
- Binary classification layer for two-class output
- Tools: PyTorch, GPU-based training

The architecture diagram:



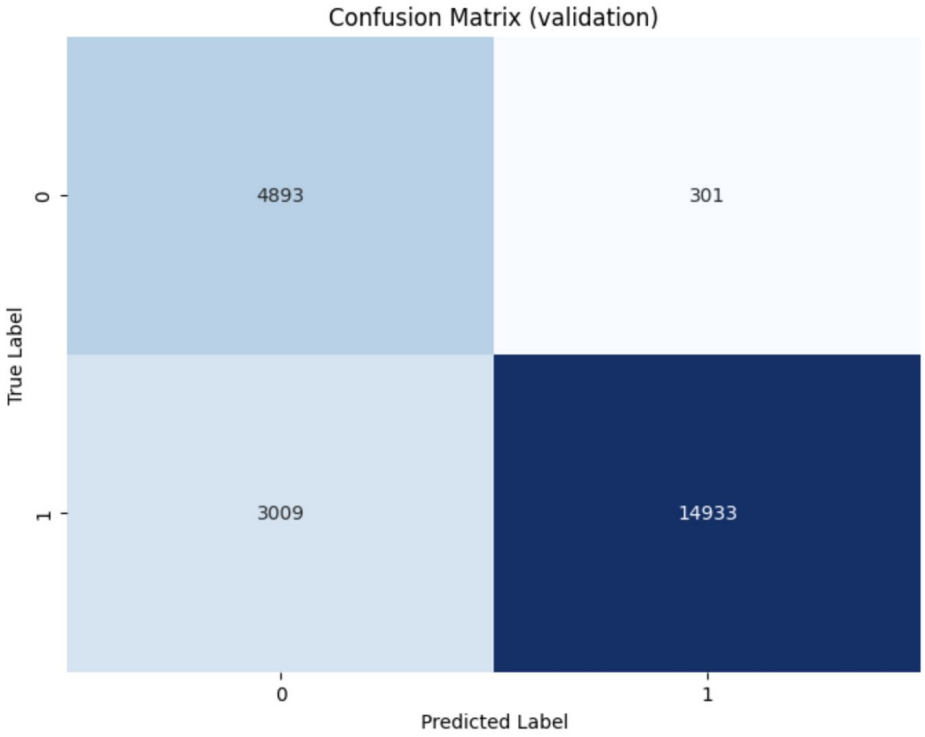
Training and Evaluation

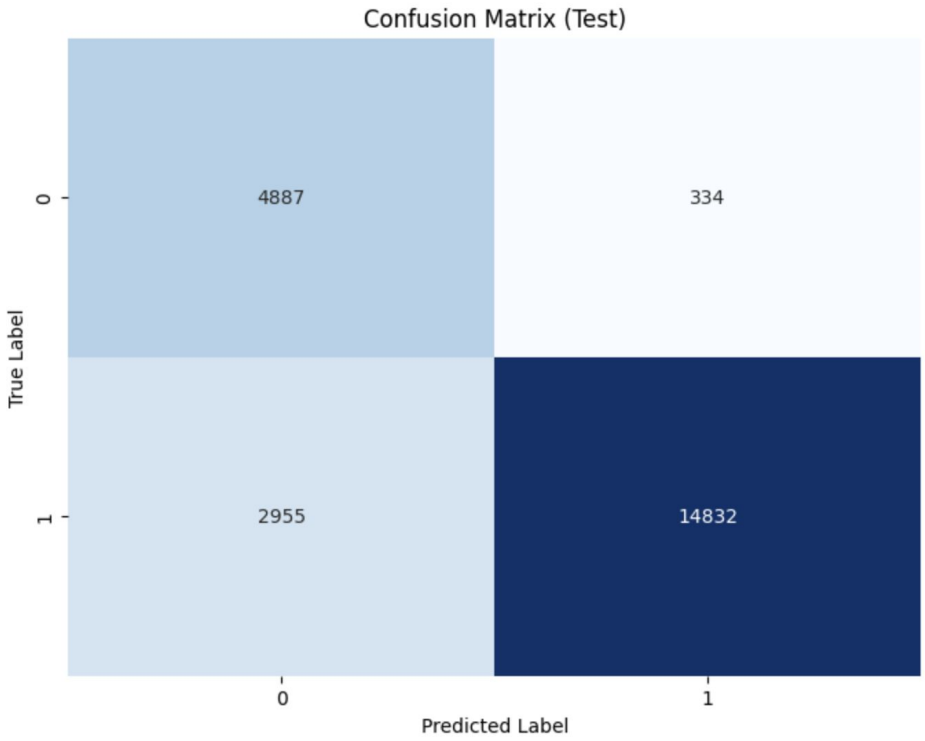
- **Training Pipeline**:
  - Custom Dataset class for image and label handling.
  - Data augmentation and preprocessing for better generalization.
  - Use of **Cross-Entropy Loss** for binary classification.
- **Evaluation**:
  - **Performance Metrics**: ROC curve, AUC (Area Under Curve), ensuring high detection sensitivity and specificity.
  - **Data Augmentation**: Applied to boost model performance in varied clinical conditions.

Achievements

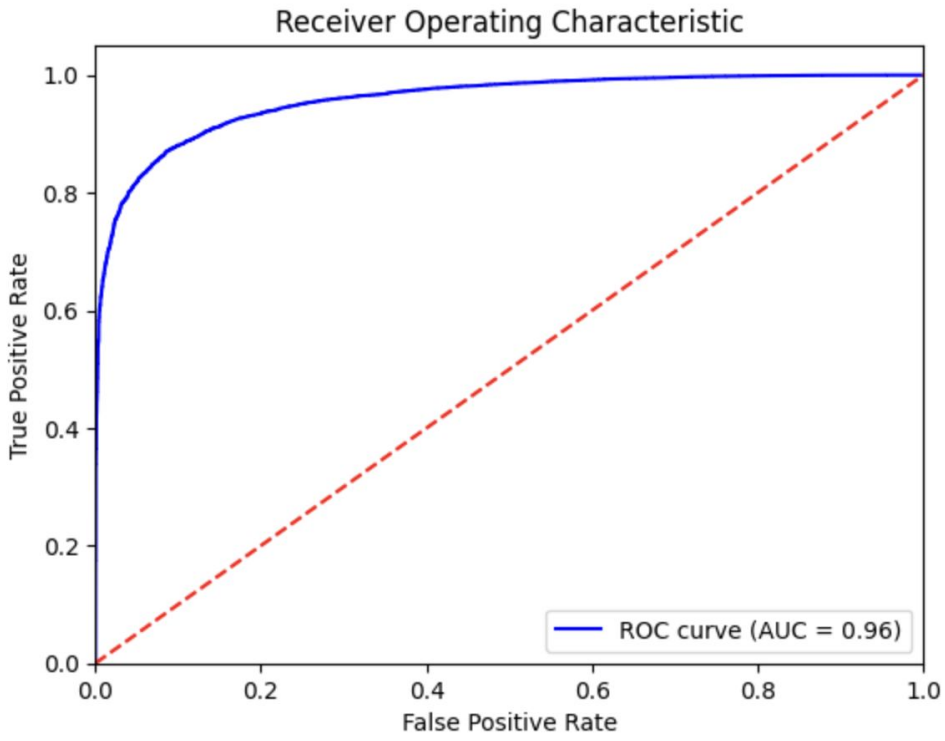
- **Business Impact**:
  - Enhanced diagnostic processes lead to better healthcare outcomes.
  - Scalable, cost-efficient AI solution integrated into pathology workflows.
  - Opened doors for clinical trials and FDA approvals for AI-driven cancer diagnostics.
- **Technical Achievements**:
  - Successful application of the Swin Transformer model to a medical dataset.
  - Improved accuracy in cancer detection through complex image analysis.
  - Demonstrated the potential for deployment in clinical settings, assisting in real-time diagnostics.

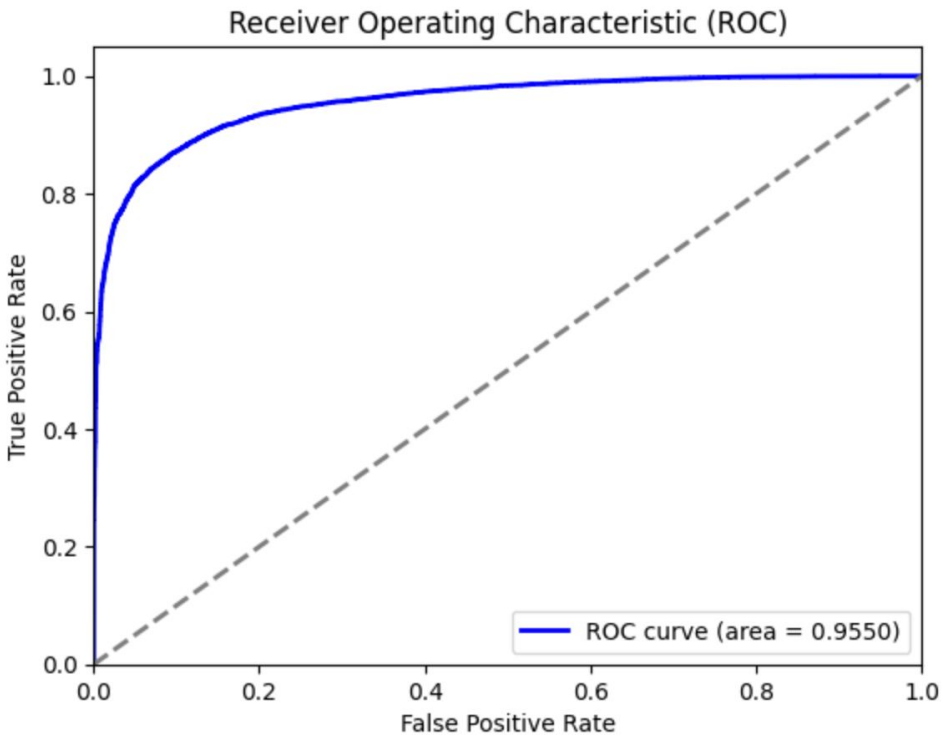
Confusion Matrices for Validation Data and Test Data:



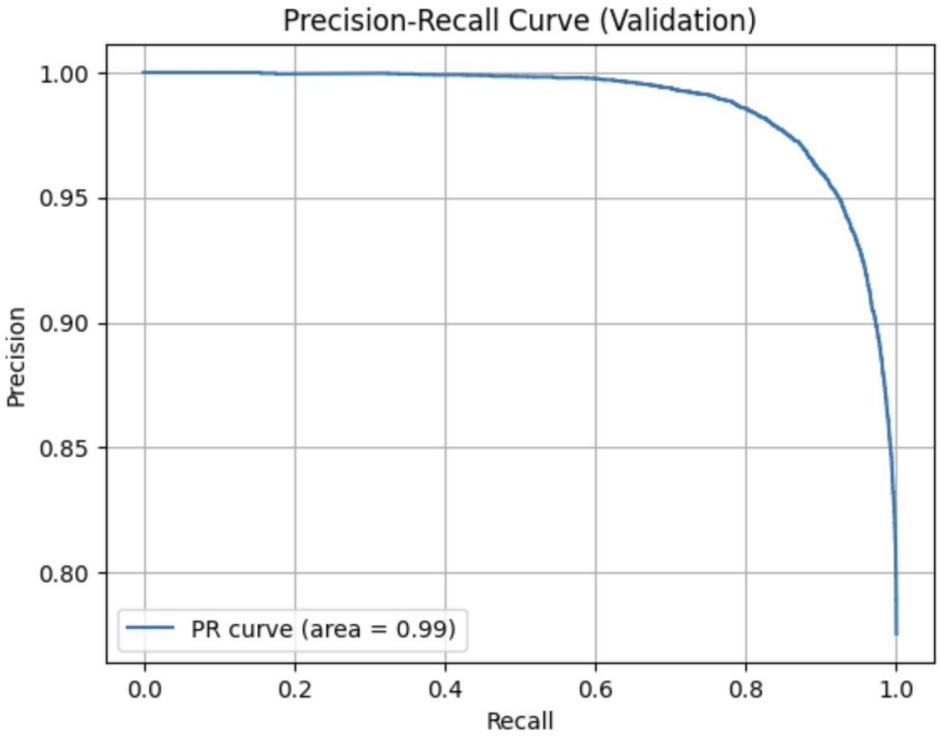


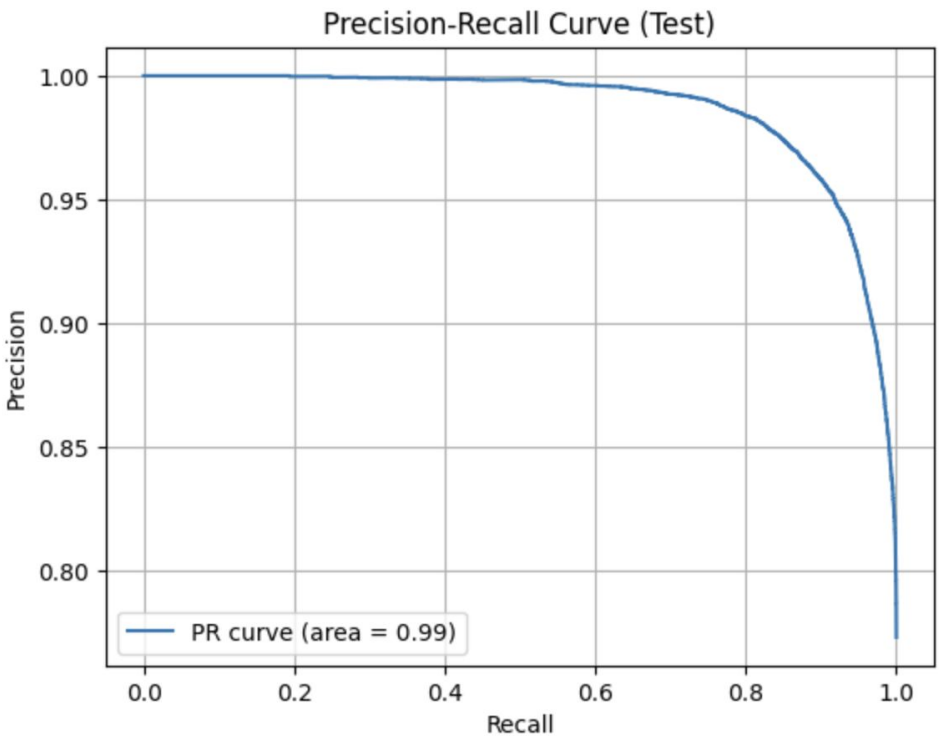
ROC AUC for Validation Data and Test Data:





Precision-Recall Curve for Validation Data and Test Data:





Histopathologic Cancer Detection: Demonstration and Conclusion

Deployment Strategy

Deployment

This app was locally deployed . The system architecture includes the following components:

- Frontend (ReactJS): Users can upload histopathologic images in tif format for analysis and view the cancer detection results. Which runs on port http://localhost:3000/

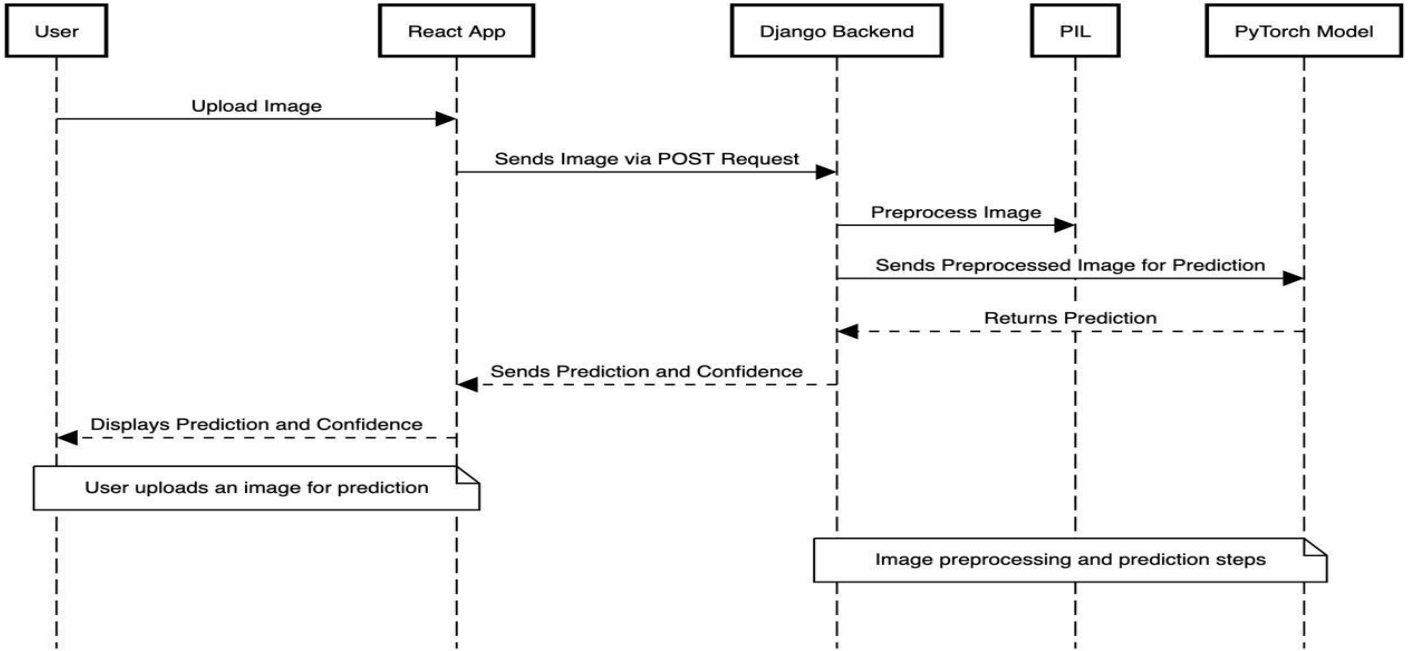
- Backend (Django): The backend processes image uploads, interacts with the machine learning model and returns the cancer prediction results. Which runs on http://127.0.0.1:8000/

Sample response :


```
'models': {
'Swin Transformer': {'is_cancerous': True, 'confidence': 0.9924},
'Custom CNN': {'is_cancerous': True, 'confidence': 0.7278}
  }
```


- Model (PyTorch): A custom CNN model trained on Kaggle's histopathologic cancer detection dataset processes the images and predicts whether they are cancerous.(mode.pth file saved in root folder)

Application workflow



Project Highlights and Future Directions

Achievements

- ROC-AUC: 0.9550
- Precision-Recall AUC: 0.9878

Follow-Up Work

- Cloud Deployment: Transition to cloud for better accessibility and scalability.
- Model Refinement: Optimize CNN and Swin Transformer with extended training.
- Explainable AI: Incorporate Grad-CAM for model decision interpretability.
- Model Ensembling: Combine multiple models to further improve accuracy

Histopathologic Cancer Detection: Team Details

Contributions

Project Manager - Ramyasri Murugesan (11810784)

Feature Engineering and Data Preprocessing Specialist - Ziya Mubeen Ahmed Mohammed (11774942)

Model Developer 1 (Approach 1) - Tarun Sadarla (11804628)

Model Developer 2 (Approach 2) - Abdul Rafay Mohd (11774936)

Deployment Specialist - Darshan Rohit Karkera (11694026)

Recording Link:

https://drive.google.com/drive/folders/1DAKgyJzvRk91gaYX1U1vtqYsDt35Ty07?usp=sharing